

RISA-bot Robotics Workshop

Day 1 Cheat Sheet & Presentation Flow

1. Environment & Setup Reference

This cheat sheet summarizes all essential commands and parameter tuning steps for Day 1 of the RISA-bot Competition Workshop. Always pay attention to whether a command is executed in the Local Laptop Terminal or inside the Robot SSH session.

Network & Connection Setup

- * **Wi-Fi:** Ensure your laptop is connected to the same Wi-Fi network as the RISA-bots.
- * **SSH Terminal:** Open a terminal on your laptop to establish an SSH connection to the robot.
- * **NoMachine:** Use NoMachine for a graphical desktop view (port 4000) to inspect camera feeds.

Host vs. Robot Commands

Context	Action	Command / Description
[LAPTOP TERMINAL]	SSH Connection	ssh sunrise@<robot_ip> (Log in with credentials; default pw is sunrise or risabot)
[LAPTOP BROWSER]	Access Dashboard	http://<robot_ip>:8080 (Web interface to adjust parameters and monitor topics live)
[ROBOT SSH]	Source Workspace	cd ~/risabotcar_ws && source install/setup.bash (Or use the alias shortcut: sos)
[ROBOT SSH]	Launch Workspace	ros2 launch risabot_automode bringup.launch.py (Launches sensors, perception, and auto-driver brain)
[ROBOT SSH]	Rebuild Workspace	cb (Alias shortcut for: colcon build --symlink-install)

2. Essential ROS 2 CLI Reference

Use these commands inside a secondary [ROBOT SSH] terminal window to monitor system state and verify topics.

Command	Context	Purpose
ros2 topic list	[ROBOT SSH]	Lists all currently active topics in the ROS graph
ros2 node list	[ROBOT SSH]	Lists all running nodes
ros2 topic echo /lane_error	[ROBOT SSH]	Streams the camera's deviation error (-1.0 to 1.0) in real-time
ros2 topic hz /scan	[ROBOT SSH]	Verifies LiDAR publishing frequency (should be ~8-12 Hz)
ros2 param get /auto_driver forward_speed	[ROBOT SSH]	Retrieves the current value of a parameter
ros2 param set /auto_driver forward_speed 0.12	[ROBOT SSH]	Modifies a parameter value immediately on the fly
ros2 topic pub --once /set_challenge std_msgs/String "data: <STATE>"	[ROBOT SSH]	Forces the state machine into specific challenge states for isolated testing (e.g. TUNNEL, LANE_FOLLOW, OBSTRUCTION)

Important Bash Aliases on Robot

Shortcuts configured in ~/.bashrc to accelerate development and debugging:

Alias	Equivalent Command	Use Case
cb	colcon build --symlink-install	Rebuild code changes
sos	source install/setup.bash	Re-source the workspace in a terminal
s	source ~/.bashrc	Reload bash configurations
run_trisabot	tmux launch scripts	Starts all nodes in a split tmux session for SSH-only operations
fix_astra	camera driver restore	Restores the Astra Mini camera driver if it goes missing

3. Parameter Tuning Reference

Modify parameters using 'ros2 param set' commands in [ROBOT SSH] or via the parameters drawer in [LAPTOP BROWSER]. Verify and lock parameters by clicking the Save (disk) button on the dashboard.

Step 1: Lane Follower (Camera Image Pipeline)

Parameter	Node	Default	Range	Description
white_threshold	line_follower_camera	200	150-240	Binary conversion threshold. Lower = more sensitive; Higher = stricter
crop_ratio_base	line_follower_camera	0.4	0.2-0.6	Crops upper portion. Higher = looks further ahead; Lower = looks in front
pid_kp	auto_driver	0.8	0.3-1.5	Proportional steering strength. Adjusts correction relative to error
pid_kd	auto_driver	0.2	0.05-0.5	Derivative steering damping. Prevents oscillations/overshooting
pid_ki	auto_driver	0.01	0.0-0.05	Integral steering drift correction. Keep low to prevent winding
forward_speed	auto_driver	0.15	0.08-0.25	Base driving speed (m/s) in straight lanes

Symptom-Fix Matrix: Lane Follower

Symptom	Probable Cause	Corrective Action
Robot oscillates/shakes left-right on straights	Proportional steering gain too high	Lower pid_kp (try 0.4 - 0.5)
Robot over-steers and wobbles after curves	Derivative damping gain too low	Increase pid_kd (try 0.3)
Robot cuts corners or hits lane borders	Looking too far ahead	Decrease crop_ratio_base (try 0.3)
Robot drifts to one side on straight tracks	System mechanical bias	Increase pid_ki (try 0.02)
Fails to detect lines (no dots on dashboard)	Detection threshold too strict	Lower white_threshold (try 150-180)
Floor textures detected as lane lines	Detection threshold too sensitive	Increase white_threshold (try 220)

Step 2: Tunnel Wall Following (LiDAR)

Parameter	Node	Default	Range	Description
kp	tunnel_wall_follower	1.2	0.5-5.0	Lateral centering strength. Too high causes wall bouncing
kd	tunnel_wall_follower	0.3	0.1-1.0	Lateral damping. Stabilizes centering corrections
kp_heading	tunnel_wall_follower	1.0	0.5-2.0	Heading alignment gain. Helps anticipate tunnel curves
min_wall_points	tunnel_wall_follower	5	2-10	Laser points needed to confirm a wall exists
target_center_dist	tunnel_wall_follower	0.0	-0.1 - 0.1	Offset adjustment if robot hugs one wall

Symptom	Probable Cause	Corrective Action
Robot oscillates back and forth in tunnel	Centering gain too high / Damping too low	Lower kp (try 3.0), raise kd (try 0.8)
Robot rides too close to one wall	Calibration/offset error	Adjust target_center_dist by + / - 0.02
Flickers in/out of tunnel mode at entrance	Too few laser scan points matched	Lower min_wall_points to 3

4. Day 1 Chronological Presentation Flow

Follow this sequence of tasks and commands step-by-step during the presentation and physical tests on Lane 1.

Step 1 ROS & Linux Recap

10:00 - 10:35 | All Groups at Station

Establish basic connection to the robot, verify system workspace status, and verify launch scripts.

```
# Step 1: Open a terminal on your laptop and SSH into the robot
[LAPTOP TERMINAL] ssh sunrise@<robot_ip>

# Step 2: Source the workspace inside the SSH session
[ROBOT SSH] cd ~/risabotcar_ws && source install/setup.bash

# Step 3: Run the system bringup launcher
[ROBOT SSH] ros2 launch risabot_automode bringup.launch.py

# Step 4: Open a second terminal, SSH in, and run verification tools
[ROBOT SSH] ros2 topic list
[ROBOT SSH] ros2 node list
[ROBOT SSH] ros2 topic echo /lane_error
[ROBOT SSH] ros2 param get /auto_driver forward_speed
```

Step 2 **RISA-bot Overview & Workshop Goals**

10:35 - 10:55 | Instructor Presentation

No coding tasks. Instructor highlights the architecture, challenge layouts, and the core pattern of using one sensor and one algorithm per challenge with automatic switching.

Step 3 **Architecture & Component Setup**

10:55 - 11:35 | All Groups at Station

Examine the launch scripts and configuration parameters on the robot. Check and test live parameter modifications.

```
# Step 1: Check the bringup launch structure
[ROBOT SSH] cat ~/risabotcar_ws/src/RISA-bot/src/risabot_automode/launch/bringup.launch.py

# Step 2: Read parameters configuration file
[ROBOT SSH] cat ~/risabotcar_ws/src/RISA-bot/src/risabot_automode/config/params.yaml

# Step 3: Set forward speed parameter live and verify
[ROBOT SSH] ros2 param set /auto_driver forward_speed 0.12
[ROBOT SSH] ros2 param get /auto_driver forward_speed

# Step 4: Revert speed to default
[ROBOT SSH] ros2 param set /auto_driver forward_speed 0.15
```

Step 4 **Dashboard Tour**

11:35 - 12:05 | All Groups at Station

Access the robot's web server dashboard from your laptop, examine debug screens, change thresholds, and save.

```
# Step 1: Open web browser on your laptop and go to the URL
[LAPTOP BROWSER] http://<robot_ip>:8080

# Step 2: Observe debug camera feed overlays. Open Parameters drawer.
# Step 3: Verify dashboard set operation inside the robot terminal
[ROBOT SSH] ros2 param get /auto_driver forward_speed

# Step 4: Tune 'white_threshold' live on dashboard (set to 50, then 220) and observe overlay changes
# Step 5: Revert to functional defaults and click Save (disk icon) on the web dashboard
```

Step 5 **Lane Detection: Image Pipeline**

12:05 - 13:00 | All Groups at Station

Tune the binary threshold and cropping parameters to isolate the white lane lines cleanly.

```
# Step 1: Set crop ratio parameter high and low to observe boundary line shifts
[ROBOT SSH] ros2 param set /line_follower_camera crop_ratio_base 0.3
[ROBOT SSH] ros2 param set /line_follower_camera crop_ratio_base 0.7

# Step 2: Set threshold parameter high and low to see boundary tracking coverage changes
[ROBOT SSH] ros2 param set /line_follower_camera white_threshold 50
[ROBOT SSH] ros2 param set /line_follower_camera white_threshold 220

# Step 3: Echo error output and slide the robot left/right to confirm error matches deviation
[ROBOT SSH] ros2 topic echo /lane_error
```

Step 6 PID Steering Tuning

14:30 - 15:10 | Lane 1 (Rotational Pairs)

Position the robot on Lane 1 curves. Start auto mode and tune the PID gains to get smooth steering without oscillation.

```
# Step 1: Set safe slow starting speeds and low gains
[ROBOT SSH] ros2 param set /auto_driver forward_speed 0.10
[ROBOT SSH] ros2 param set /auto_driver pid_kp 0.5
[ROBOT SSH] ros2 param set /auto_driver pid_kd 0.2
[ROBOT SSH] ros2 param set /auto_driver pid_ki 0.01

# Step 2: Press Start on gamepad controller to enable auto mode
# Step 3: If robot oscillates, raise derivative damping gain
[ROBOT SSH] ros2 param set /auto_driver pid_kd 0.3
# Step 4: If robot cuts corners, lower crop ratio
[ROBOT SSH] ros2 param set /line_follower_camera crop_ratio_base 0.3
```

Step 7 Tunnel Navigation (LiDAR)

15:10 - 16:10 | Lane 1 (Rotational Pairs)

Test LiDAR wall tracking and verify auto mode transitions at the tunnel entrance/exit.

```
# Step 1: Verify wall measurements in secondary terminal
[ROBOT SSH] ros2 topic echo /tunnel_debug

# Step 2: Set initial tunnel parameters
[ROBOT SSH] ros2 param set /tunnel_wall_follower forward_speed 0.10
[ROBOT SSH] ros2 param set /tunnel_wall_follower kp 4.0
[ROBOT SSH] ros2 param set /tunnel_wall_follower kd 0.6

# Step 3: Press Start on gamepad to run from lane follower -> tunnel -> exit.
# Step 4: Monitor topics in separate terminals to verify switching
[ROBOT SSH] ros2 topic echo /lane_error
[ROBOT SSH] ros2 topic echo /tunnel_cmd_vel
```

Step 8 Full Lane 1 Tuning Run

16:10 - 17:00 | Lane 1 (Rotational Pairs)

Verify end-to-end performance and save the final values permanently.

```
# Step 1: Set finalized speed and gain parameters
[ROBOT SSH] ros2 param set /auto_driver forward_speed 0.15
[ROBOT SSH] ros2 param set /auto_driver pid_kp 0.8
[ROBOT SSH] ros2 param set /auto_driver pid_kd 0.20
[ROBOT SSH] ros2 param set /line_follower_camera white_threshold 100
[ROBOT SSH] ros2 param set /line_follower_camera crop_ratio_base 0.55
[ROBOT SSH] ros2 param set /tunnel_wall_follower kp 5.0
[ROBOT SSH] ros2 param set /tunnel_wall_follower kd 0.5

# Step 2: Run the full course. Verify transitions.
# Step 3: Click Save on the Dashboard to save parameters to
~/risabotcar_ws/src/RISA-bot/src/risabot_automode/config/params.yaml
```

Day 1 Tuning Sequence Rule (Never Skip)

Always tune in the following sequence during practice and competition:

- * 1. Clean binary camera detection (white_threshold).
- * 2. Target image crop window (crop_ratio_base).
- * 3. Stable camera-based steering gains (pid_kp, pid_kd).

- * 4. Stable LiDAR wall-following gains (tunnel kp, kd).
- * 5. Increase forward speed (forward_speed) incrementally last.